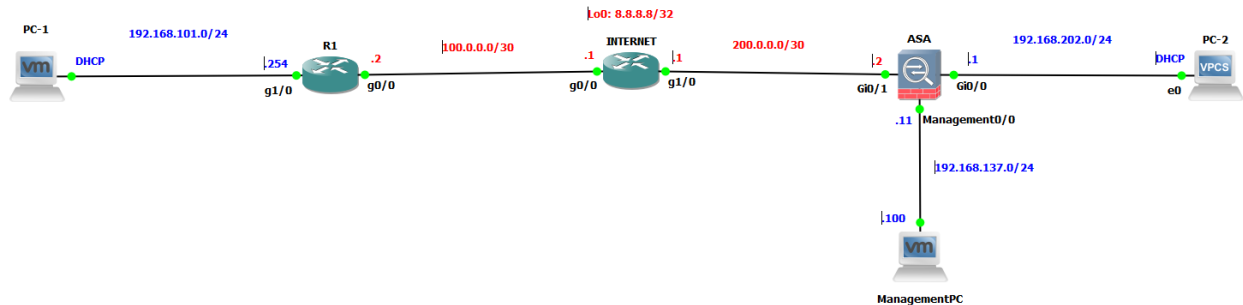


Cấu hình Site-to-Site VPN trên ASA bằng ASDM

Sơ đồ



Yêu cầu

- Cấu hình cơ bản :
 - Đặt IP cho các thiết bị
 - Cấu hình R1 làm DHCP Server để PC-1 có thể nhập IP bằng DHCP
 - Cấu hình định tuyến và NAT để PC-1 có thể truy cập Internet
 - Cấu hình để có thể quản trị hai ASA bằng ASDM
 - Cấu hình các tính năng cơ bản cho ASA trên ASDM như : IP, DHCP, định tuyến, NAT, và policy để PC-2 có thể nhận IP bằng DHCP và truy cập Internet
- Cấu hình Remote Access VPN trên ASA để PC-1 có thể truy cập vào mạng nội bộ và trao đổi dữ liệu với PC-2

Các bước thực hiện

Bước 1 : Cấu hình cơ bản cho các thiết bị

1.1. Cấu hình Router INTERNET

```
INTERNET (config) # interface Gi0/0
INTERNET (config-if) # ip address 100.0.0.1 255.255.255.252
INTERNET (config-if) # no shutdown
INTERNET (config-if) # exit
INTERNET (config) # interface Gi1/0
INTERNET (config-if) # ip add 200.0.0.1 255.255.255.252
INTERNET (config-if) # no shutdown
INTERNET (config-if) # exit
INTERNET (config) # interface loopback0
INTERNET (config-if) # ip address 8.8.8.8 255.255.255.255
INTERNET (config-if) # exit
```

1.2. Cấu hình Router R1

```
R1(config)# interface Gi0/0
R1(config-if)# ip address 100.0.0.2 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface Gi1/0
R1(config-if)# ip address 192.168.101.254 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# ip dhcp pool LAN
R1(dhcp-config)# network 192.168.101.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.101.254
R1(dhcp-config)# dns 8.8.8.8
R1(dhcp-config)# exit

R1(config)# ip route 0.0.0.0 0.0.0.0 100.0.0.1

R1(config)# access-list 1 permit
R1(config)# ip nat inside source list 1 interface Gi0/0 overload
R1(config)# interface Gi0/0
R1(config-if)# ip nat outside
R1(config-if)# exit
R1(config)# interface Gi1/0
R1(config-if)# ip nat inside
```

1.3. Cấu hình cơ bản để quản trị ASA bằng ASDM

```
ASA(config)# interface management 0/0
ASA(config-if)# nameif mgmt
ASA(config-if)# security-level 100
ASA(config-if)# ip address 192.168.137.11 255.255.255.0
ASA(config-if)# no shutdown
ASA(config-if)# exit

ASA(config)# username admin password cisco

ASA(config)# http server enable
ASA(config)# http 192.168.137.0 255.255.255.0 mgmt
```

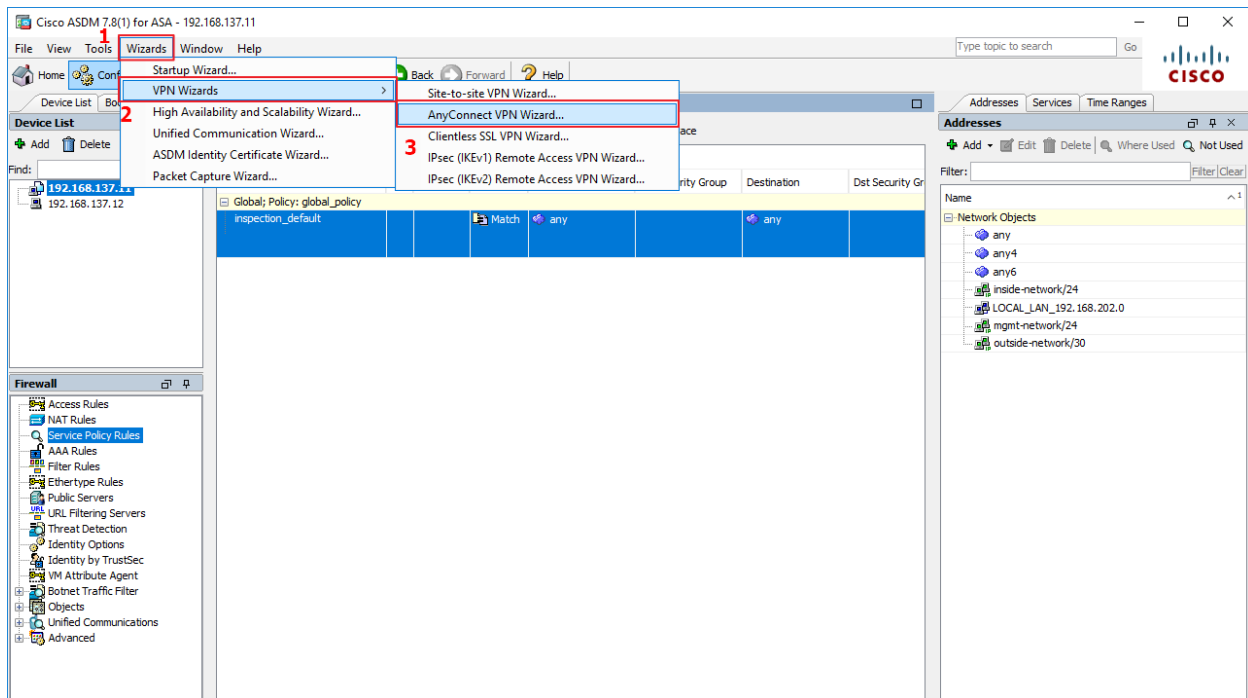
Cấu hình các tính năng cơ bản trên ASA vui lòng tham khảo bài lab cấu hình cơ bản.

Các menu cần nhớ :

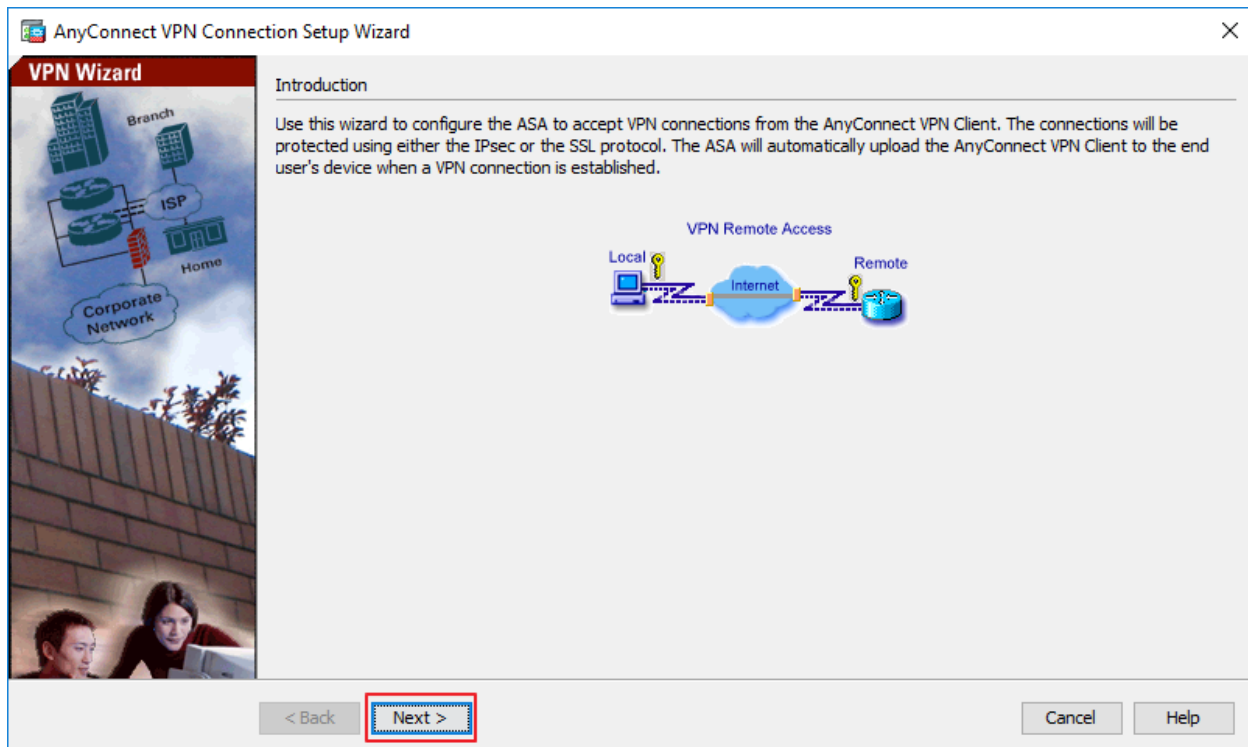
- Cấu hình IP Interface : Configuration > Device Setup > Interface Settings > Interfaces
- Cấu hình DHCP : Configuration > Device Management > DHCP > DHCP Server
- Cấu hình định tuyến tĩnh : Configuration > Device Setup > Routing > Static Route
- Cấu hình NAT : Configuration > Firewall > NAT Rules
- Cấu hình Policy : : Configuration > Firewall > Service Policy Rules

Bước 2 : Cấu hình Remote Access VPN trên ASA bằng ASDM

Vào menu **Wizards > VPN Wizards > AnyConnect VPN Wizard...**



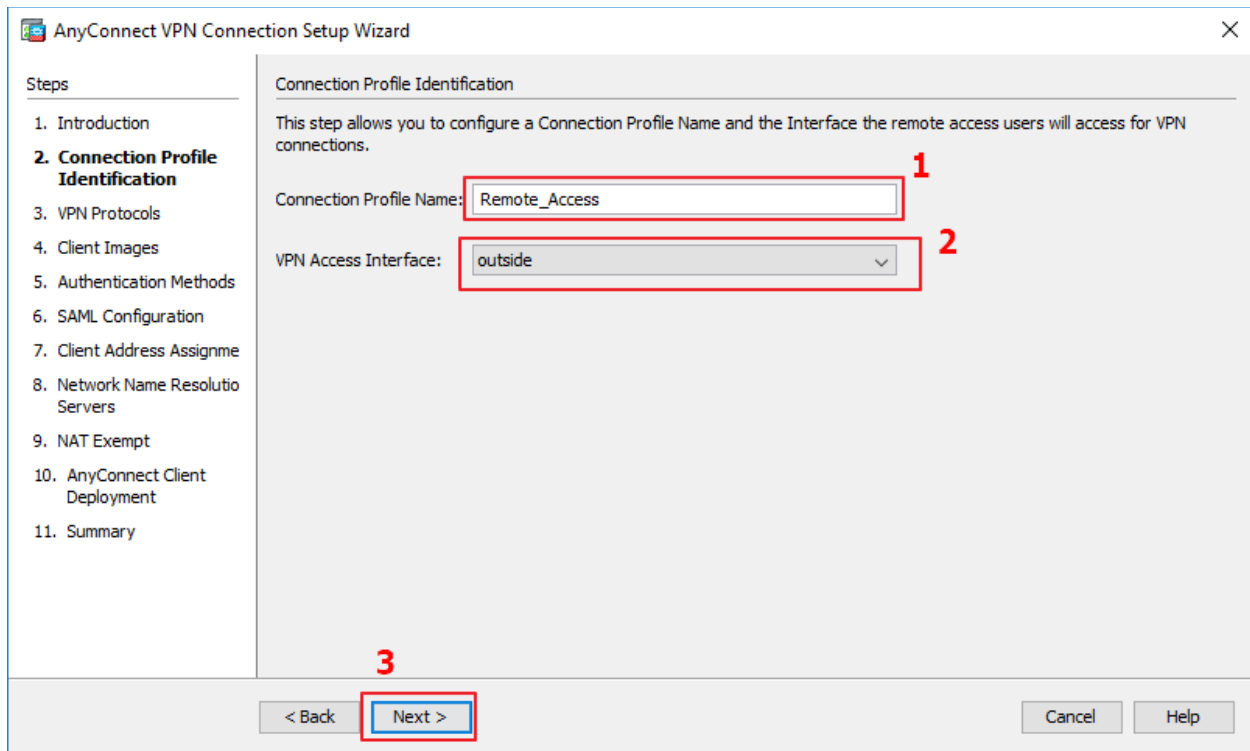
Click **Next**



Đặt tên **Profile** (tên profile có thể đặt tùy ý)

Chọn **VPN Access Interface** là **outside**.

Click **Next**

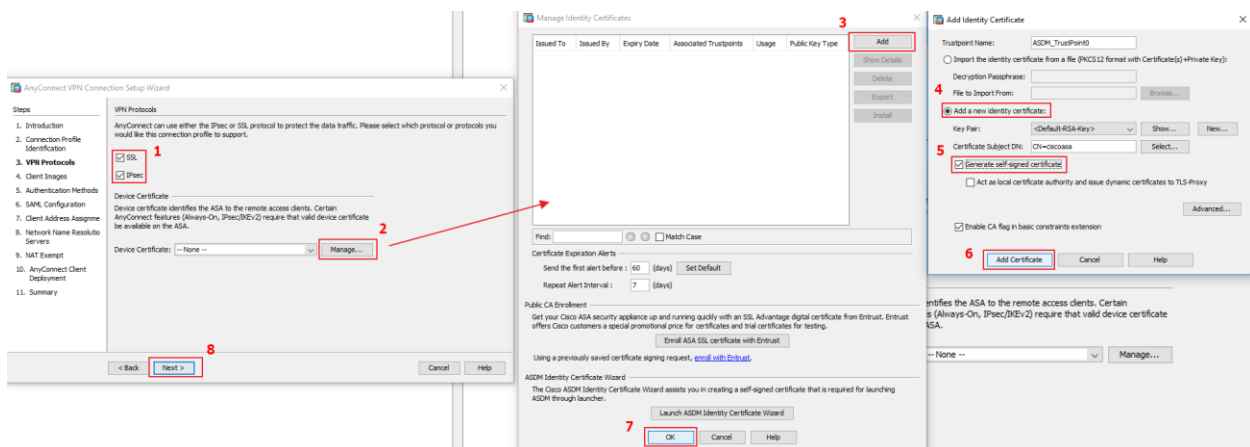


The screenshot shows the 'AnyConnect VPN Connection Setup Wizard' window, specifically the 'Connection Profile Identification' step. On the left, a 'Steps' list shows '2. Connection Profile Identification' as the current step. The main area contains two fields: 'Connection Profile Name' with the value 'Remote_Access' (highlighted with a red box and number 1), and 'VPN Access Interface' with the value 'outside' (highlighted with a red box and number 2). At the bottom, the 'Next >' button is highlighted with a red box and number 3. 'Back' and 'Cancel' buttons are also visible.

Check vào **SSL** và **IPSEC**

Click **Manage** ở mục **Device Certificate** (Nếu ASA đã có Certificate thì không cần làm phần này). Click **Add**, tick chọn **Add a new identity certificate**. Tick vào **generate self-signed certificate**. Click **Add Certificate**. Click **OK**

Click **Next**



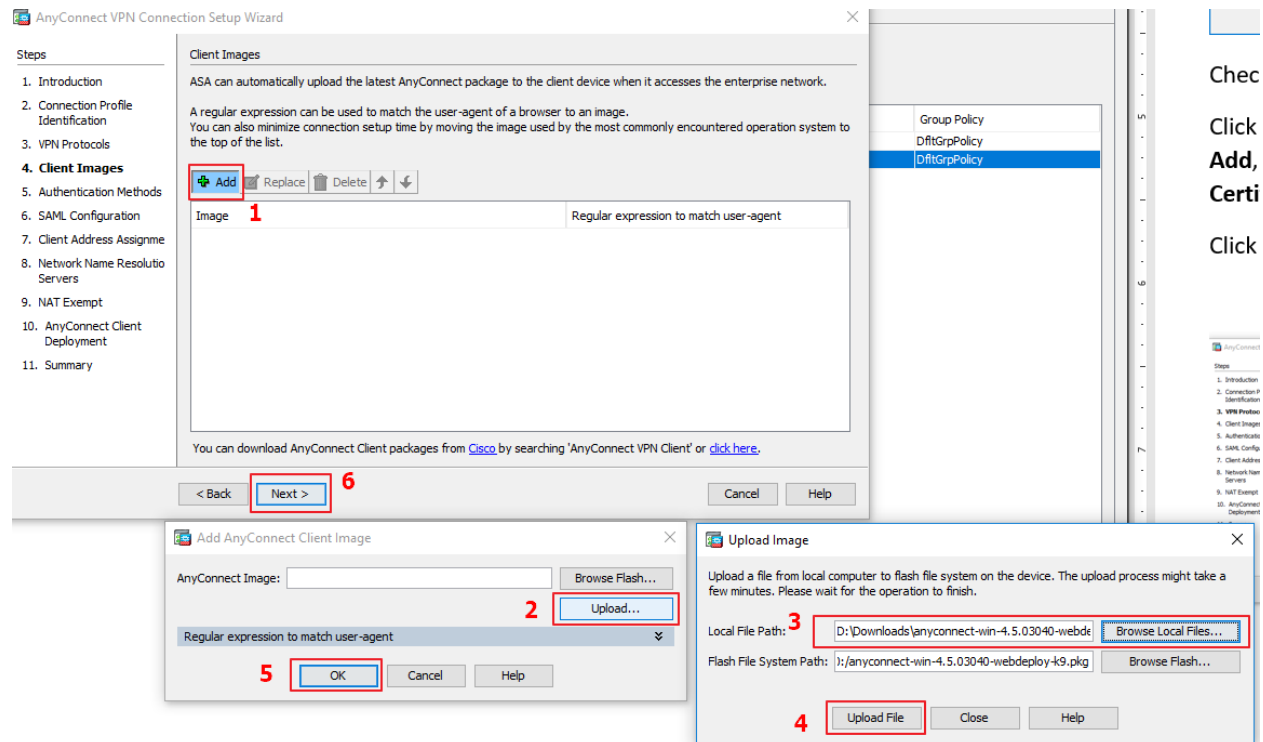
This block contains three overlapping screenshots illustrating the configuration steps for a Device Certificate. The leftmost screenshot shows the 'VPN Protocols' section with 'SSL' and 'IPsec' checked (number 1), and the 'Device Certificate' dropdown set to 'None' with a 'Manage...' button highlighted (number 2). The middle screenshot shows the 'Manage Identity Certificates' dialog with the 'Add' button highlighted (number 3). The rightmost screenshot shows the 'Add Identity Certificate' dialog with 'Add a new identity certificate' selected (number 4), 'Generate self-signed certificate' checked (number 5), and the 'Add Certificate' button highlighted (number 6). A 'Next >' button from the first screenshot is also highlighted with a red box and number 8.

Click **Add**

Click **Upload** (nếu ASA đã có image Anyconnect thì click **Browse Flash**). Click chọn **Browse Local File** và upload file **anyconnect-win-4.5.03040-webdeploy-k9.pkg**. Click **Upload File**

Click **OK**

Click **Next**



Chọn **AAA Server Group** là **LOCAL**

Tạo thêm tài khoản để VPN (sau khi nhập username/password, click **Add** để thêm, click **Delete** để xóa tài khoản)

Click **Next**

AnyConnect VPN Connection Setup Wizard

Steps

1. Introduction
2. Connection Profile Identification
3. VPN Protocols
4. Client Images
- 5. Authentication Methods**
6. SAML Configuration
7. Client Address Assignment
8. Network Name Resolution Servers
9. NAT Exempt
10. AnyConnect Client Deployment
11. Summary

Authentication Methods

This step lets you specify the location of the authentication server. You can click on the "New..." button to create a new server group.

AAA Server Group: LOCAL New...

Local User Database Details

User to be Added

Username: thinhnp

Password:

Confirm Password:

Add >> Delete

admin

< Back Next > Cancel Help

Click **Next**

AnyConnect VPN Connection Setup Wizard

Steps

1. Introduction
2. Connection Profile Identification
3. VPN Protocols
4. Client Images
5. Authentication Methods
- 6. SAML Configuration**
7. Client Address Assignment
8. Network Name Resolution Servers
9. NAT Exempt
10. AnyConnect Client Deployment
11. Summary

SAML Configuration

This step allows you to configure a SAML and the authentication method.

Authentication

Method: AAA

AAA Server Group: LOCAL Manage...

☐ Use LOCAL if Server Group fails

SAML Identity Provider

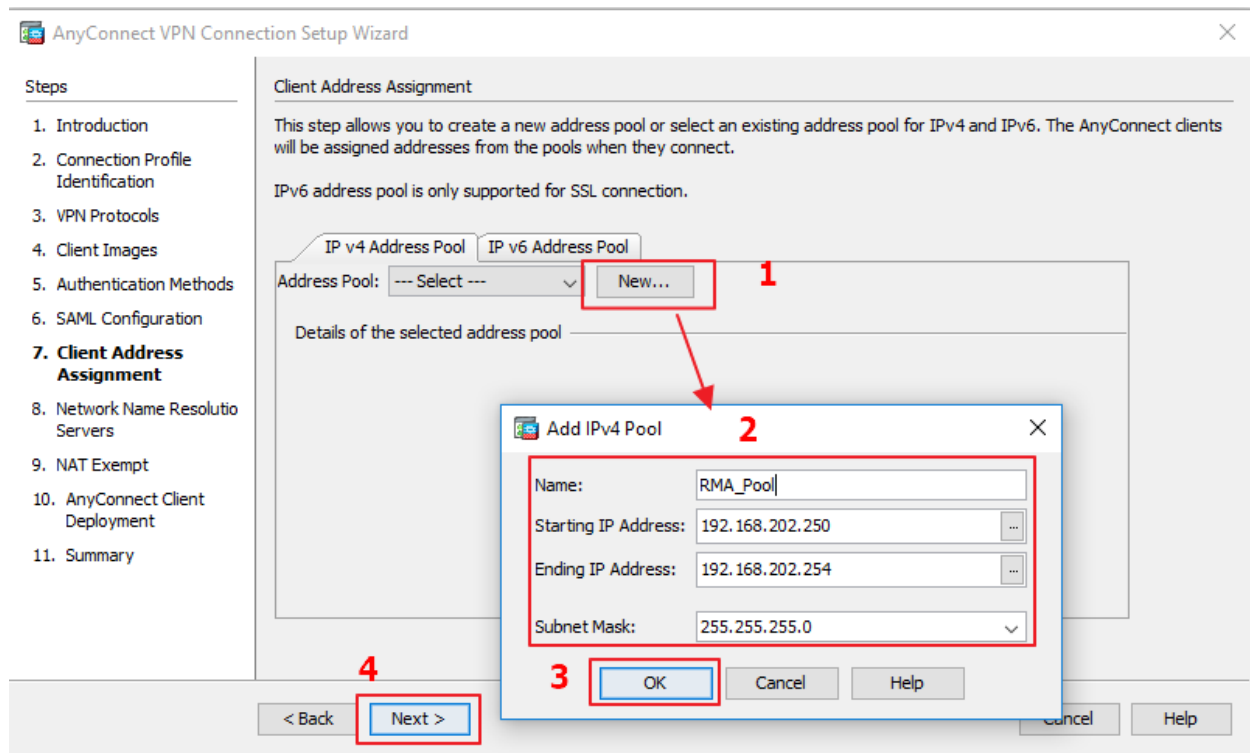
SAML Server : --- None --- Manage...

< Back Next > Cancel Help

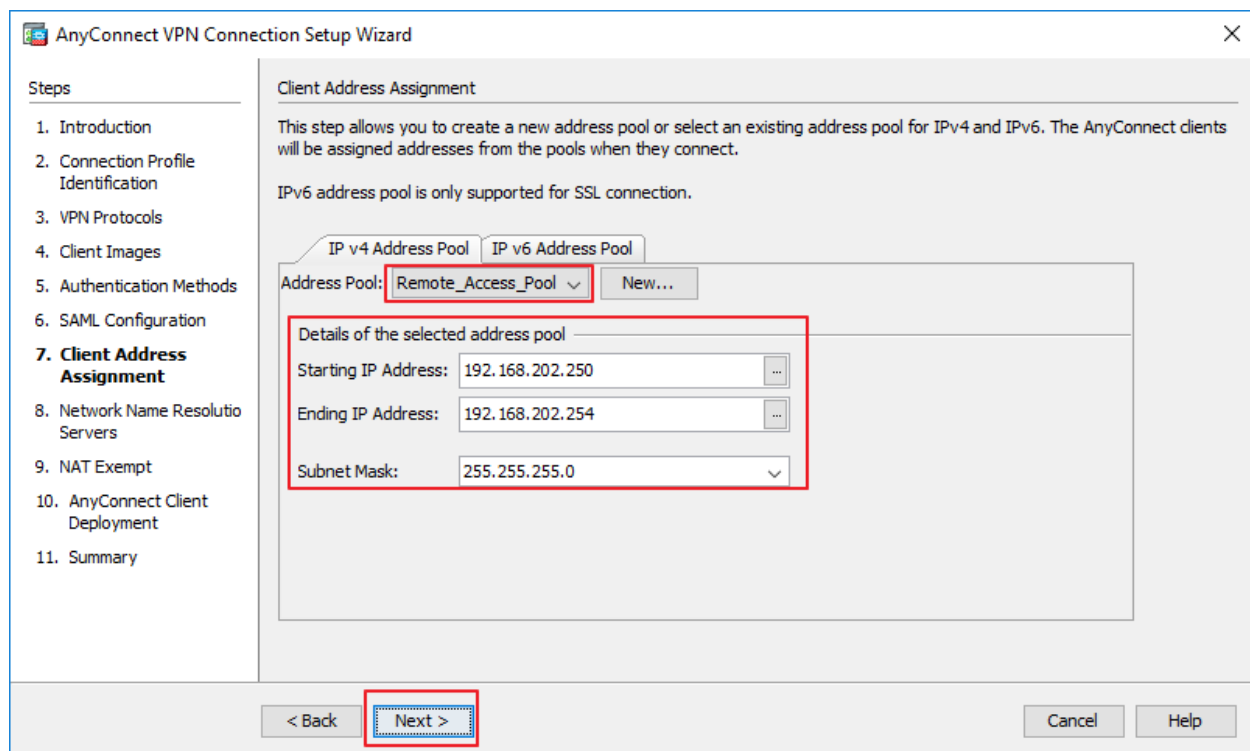
Click New ở mục **Address Pool**

Đặt tên cho pool, chọn dải IP để cấp cho client khi VPN. Click **OK**

Click **Next**

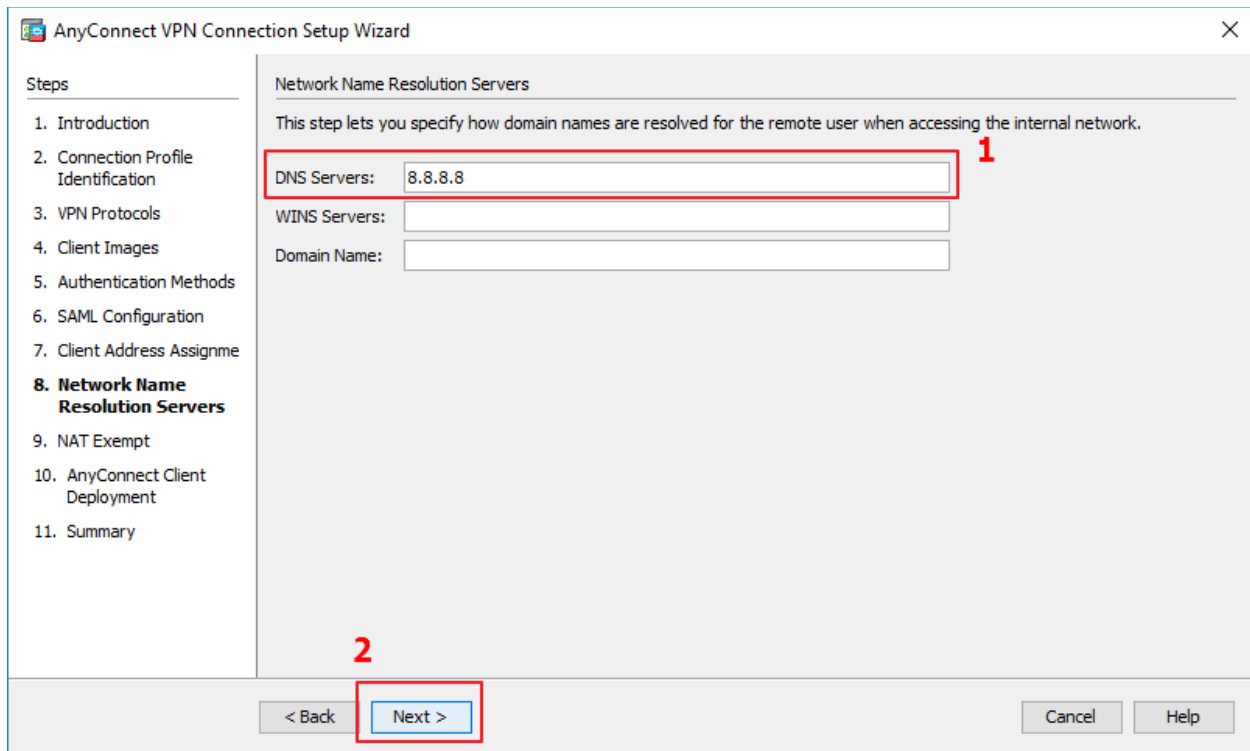


Nếu đã có pool trước đó thì không cần tạo lại. Lúc này chỉ cần chọn pool đã tạo trước đó, kiểm tra dải IP và click Next



Đặt DNS Server là 8.8.8.8

Click **Next**



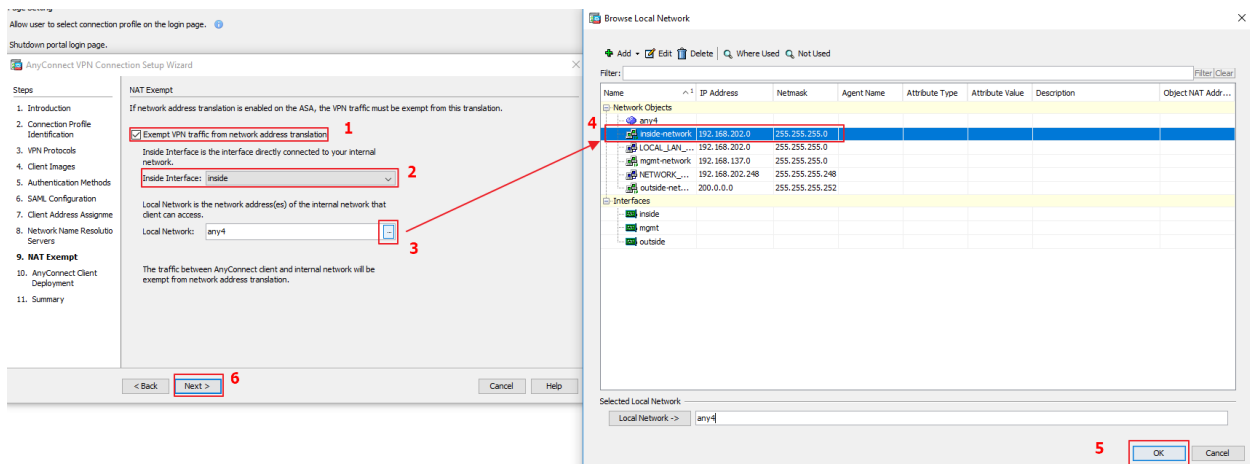
The screenshot shows the 'AnyConnect VPN Connection Setup Wizard' window, specifically the 'Network Name Resolution Servers' step. The left sidebar lists steps from 1 to 11, with step 8, 'Network Name Resolution Servers', highlighted. The main area contains the following text: 'This step lets you specify how domain names are resolved for the remote user when accessing the internal network.' Below this text are three input fields: 'DNS Servers:' with the value '8.8.8.8', 'WINS Servers:', and 'Domain Name:'. A red box highlights the 'DNS Servers' field, and a red number '1' is placed next to it. At the bottom of the window, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a red box, and a red number '2' is placed next to it.

Tick vào mục **Exempt VPN traffic from network address translation**

Chọn **Inside Interface** là **inside**

Click vào nút “...” ở mục Local Network, chọn object inside-network. Chọn Object inside-network và click OK

Click Next



The screenshot shows the 'AnyConnect VPN Connection Setup Wizard' window, specifically the 'NAT Exempt' step. The left sidebar lists steps from 1 to 11, with step 9, 'NAT Exempt', highlighted. The main area contains the following text: 'If network address translation is enabled on the ASA, the VPN traffic must be exempt from this translation.' Below this text are three input fields: 'Exempt VPN traffic from network address translation' (checked), 'Inside Interface: inside', and 'Local Network: any4'. A red box highlights the 'Exempt VPN traffic from network address translation' checkbox, and a red number '1' is placed next to it. A red box highlights the 'Inside Interface: inside' dropdown menu, and a red number '2' is placed next to it. A red box highlights the 'Local Network: any4' dropdown menu, and a red number '3' is placed next to it. At the bottom of the window, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a red box, and a red number '6' is placed next to it. A red arrow points from the 'Local Network: any4' dropdown menu to the 'Browse Local Network' dialog box. The 'Browse Local Network' dialog box shows a table of network objects. The table has columns: Name, IP Address, Netmask, Agent Name, Attribute Type, Attribute Value, Description, and Object NAT Addr... The table contains the following data:

Name	IP Address	Netmask	Agent Name	Attribute Type	Attribute Value	Description	Object NAT Addr...
any4							
any4							
LOCAL_LAN...	192.168.202.0	255.255.255.0					
mgmt-network	192.168.137.0	255.255.255.0					
NETWORK...	192.168.202.248	255.255.255.248					
outside-net...	200.0.0.0	255.255.255.252					
Interfaces							
inside							
mgmt							
outside							

A red box highlights the 'LOCAL_LAN...' row, and a red number '4' is placed next to it. At the bottom of the dialog box, there are two buttons: 'OK' and 'Cancel'. The 'OK' button is highlighted with a red box, and a red number '5' is placed next to it.

Click **Next**

AnyConnect VPN Connection Setup Wizard

Steps

1. Introduction
2. Connection Profile Identification
3. VPN Protocols
4. Client Images
5. Authentication Methods
6. SAML Configuration
7. Client Address Assignment
8. Network Name Resolution Servers
9. NAT Exempt
- 10. AnyConnect Client Deployment**
11. Summary

AnyConnect Client Deployment

AnyConnect client program can be installed to a client device by one of the following two methods:

- 1) Web launch - On accessing the ASA using a Web Browser, the AnyConnect client package will be automatically installed;
- 2) Pre-deployment - Manually install the AnyConnect client package.

☒ Allow Web Launch

Caution: Web launch is global setting, it affects all connections. AnyConnect SSL connections and clientless SSL connections will NOT work if it is turned off.

For pre-deployment, please remember to include the client profile 'disk0:/Remote_Access_client_profile.xml' from the ASA in your AnyConnect package, otherwise IPsec connection will NOT work.

< Back **Next >** Cancel Help

Click **Finish**

AnyConnect VPN Connection Setup Wizard

VPN Wizard

Summary

Here is the summary of the configuration.

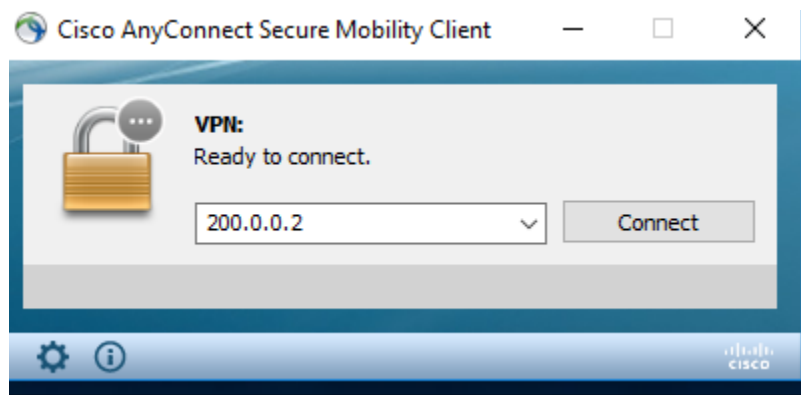
Name	Value
Summary	
Name/Alias of the Connection Profile	Remote_Access
VPN Access Interface	outside
Device Digital Certificate	ASDM_TrustPoint0:hostname=ciscoasa, cn=ciscoasa:ho...
VPN Protocols Enabled	IPsec and SSL
AnyConnect Client Images	1 package
Authentication Server Group	LOCAL
Address Pool for the Client	192.168.202.250 - 192.168.202.254
DNS	Server: Domain Name:
Network Address Translation	The protected traffic is not subjected to network address translation

< Back **Finish** Cancel Help

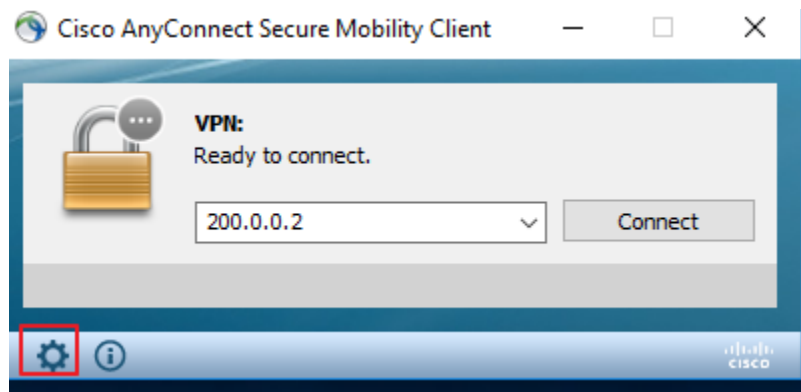
Bước 3 : Kết nối remote access bằng Cisco Anyconnect

Trên PC-1, khởi động phần mềm **Cisco Anyconnect Mobility Client**

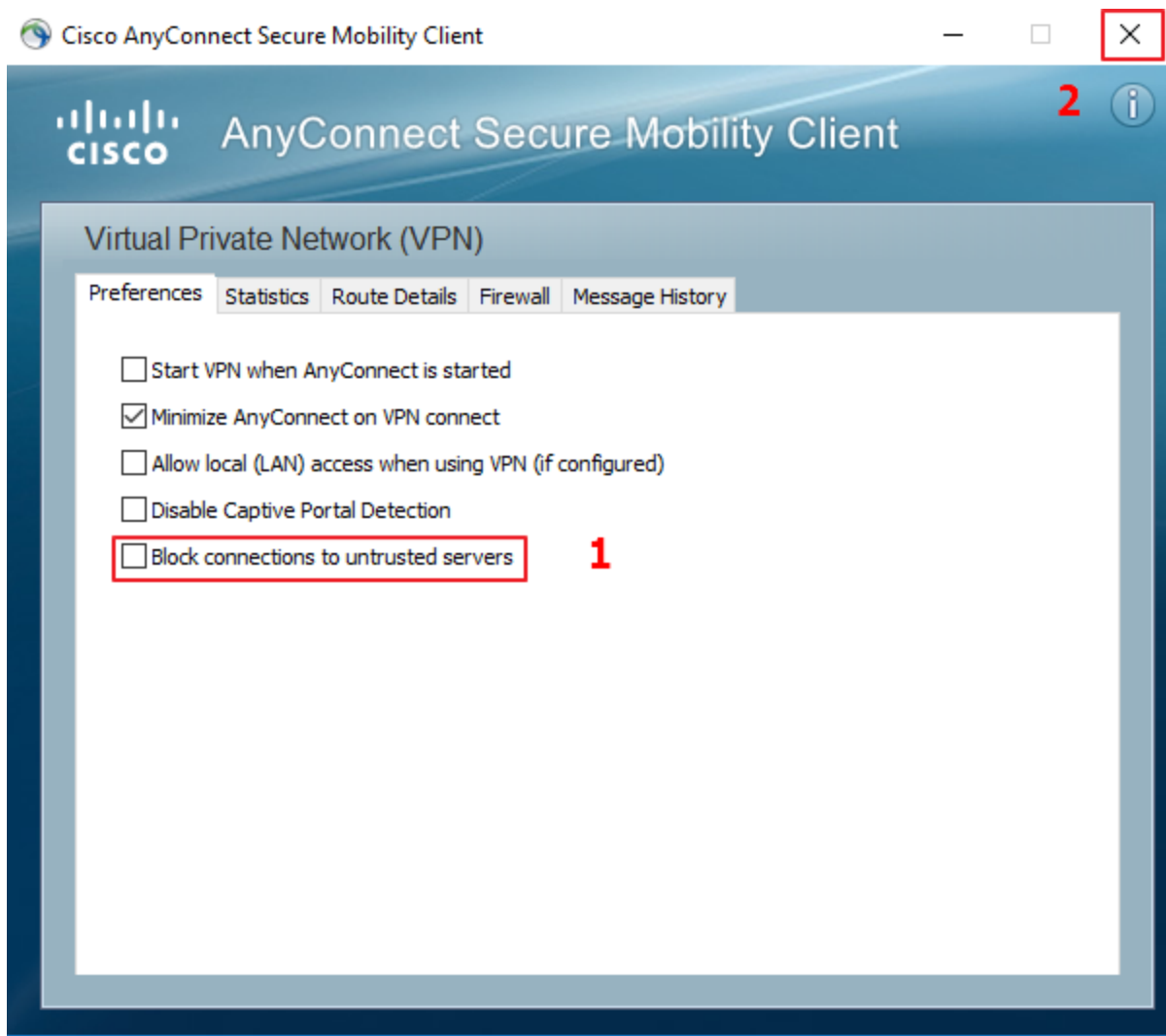
Giao diện sau khi khởi động :



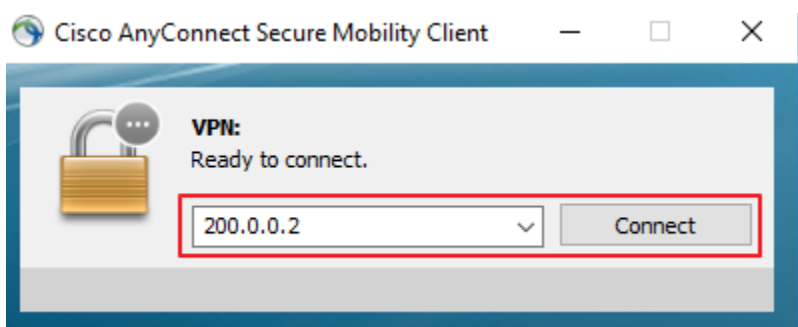
Click vào nút **Settings**



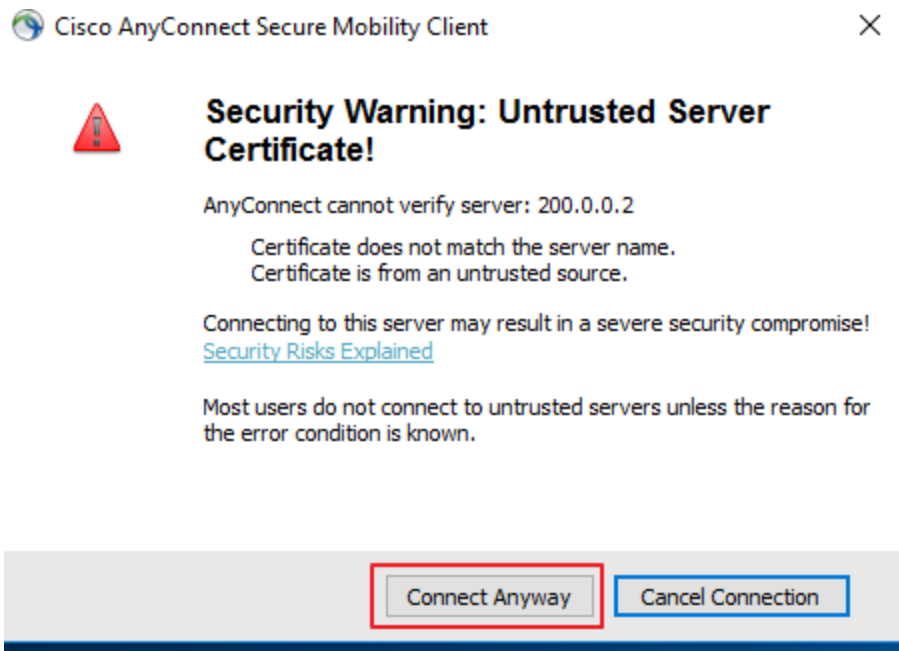
Bỏ chọn mục **“Block connection from untrusted servers”**, sau đó đóng cửa sổ



Trở lại giao diện của Anyconnect, nhập **IP public** của ASA và click **Connect**

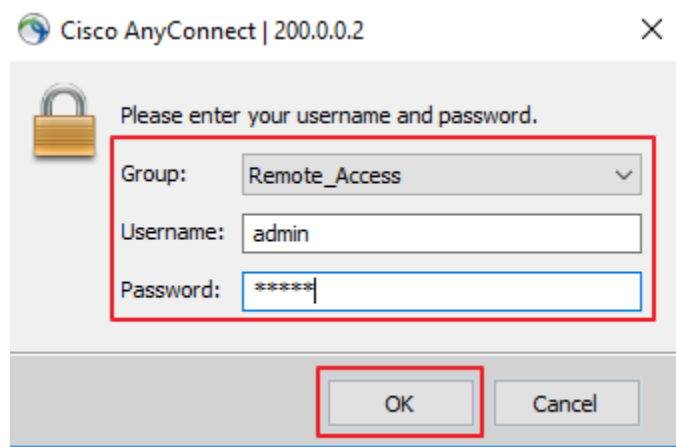


Click **Connect Anyway**

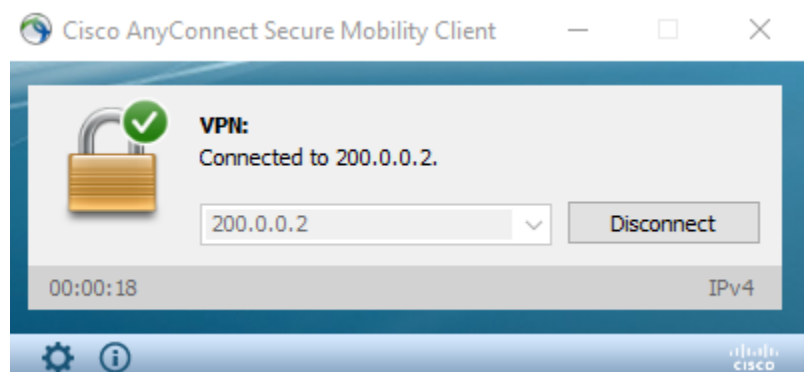


Chọn Group **Remote_Access** (là **profile name** đã tạo ở bước 2).

Nhập **username/password** và click **OK**



Giao diện khi kết nối thành công



Lúc này, trên cmd của PC-1, nếu gõ ipconfig, có thể thấy PC-1 có thêm 1 interface.

Interface này chính là interface Tunnel để kết nối về mạng nội bộ của PC-2

```
Command Prompt

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::4c82:54ec:6cec:873d%23
    Link-local IPv6 Address . . . . . : fe80::6822:b256:1354:6881%23
    IPv4 Address. . . . . : 192.168.202.250
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : ::
                               192.168.202.1

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1b7:afee:691:439%10
    IPv4 Address. . . . . : 192.168.101.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.101.254

Tunnel adapter isatap.{23D8B78B-FC6E-486E-97D9-AD3A9F3AC861}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{942B7FAF-D48C-40C5-9EF1-A19A59CF5DF3}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\admin>
```

IP/subnet mask và default gateway nằm trong dải IP đã cấp ở bước 2.

Lúc này, PC-1 có thể ping PC-2

```
Command Prompt

Microsoft Windows [Version 10.0.10586]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Users\admin>ping 192.168.202.2

Pinging 192.168.202.2 with 32 bytes of data:
Reply from 192.168.202.2: bytes=32 time=41ms TTL=64
Reply from 192.168.202.2: bytes=32 time=39ms TTL=64
Reply from 192.168.202.2: bytes=32 time=36ms TTL=64
Reply from 192.168.202.2: bytes=32 time=41ms TTL=64

Ping statistics for 192.168.202.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 41ms, Average = 39ms

C:\Users\admin>
```